Clay Mineral Structure

Video 8.1

Virtual Museum of Soils
https://chem.beloit.edu/edetc/pmks/index.htm

Oxide Clays

- Si - Quartz SiO_2
- Al - Gibbsite $Al(OH)_3$
- Fe - Ferrihydrite $Fe(OH)_3$
- Mg - Brucite $Mg(OH)_2$

Remember the Charges

- Si +4 – small (tetrahedral
- Al +3 – med. (tet or octal
- Fe +2 or +3 – large (Mos
- Mg +3 - large (Mostly oct

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Deciphering the differences

- Formulas are written as
  - $K(Mg,Fe)_3$ $[Si_3AlO_{10}]$ $(OH)_2$
  - Interlayer cations
  - Octahedral cations
  - Tetrahedral cations
- Compare the ratio of Octahed. to Tetrahed. cations

Layer Silicate Clays

- 1:1 Si:Al sheets
  - Kaolinite $Al_2Si_20_5(OH$
  - Lizardite $Mg_3Si_20_5(O$
- 2:1 Si:Al Sheets
  - Pyrophyllite $Al_2Si_40_{10}($
  - Talc $Mg_3Si_40_{10}(OH)_2$

Isomorphic Substitution: Clay Minerals

Substitution in Si Layer

- Muscovite $KAl_2(AlSi_3)O_{10}(OH,F)_2$
- Vermiculite $(Mg_3)(Al,Si)_4O_{10}(OH)_2 \cdot 4H_2O$

Substitution in Al Layer

- Montmorilonite
- Bentonite
- Soil Smectite
- $(Al_{2/3}Fe_{2/9}Mg_{1/9})Si_4O_{10}($ $\cdot 4H_2O$

Isomorphic Substitution: Clay Minerals

Substitution in SI & Al layers

- Biotite
  $K(Mg,Fe)_3AlSi_3O_{10}(OH)_2$

2:1:1 Layer Silicate

- Chlorite
  - 2 Si tetrahedral sheet
  - 1 Al Octahedra sheet
  - 1 Mg Octahedra sheet
    - Hydrated interlayer cati
      Mg,
    - Crystaline structure = b
    - $[(Mg,FeII)(OH)_2].[Si_{4-x}Al_{y}$
      $_{y}(Al,Fe)y]O_{10}(OH)_2$

Summary - Clay minerals are differentiated

Number of sheets

- 1:1
  - Kaolinite
  - Lizardite
- 2:1
  - Pyrophyllite
  - Talc
  - Biotite
  - Vermiculite
- 2:1:1
  - Chlorite

Di- Tri-octahedral

- Dioctahedral - Al
  - Muscovite
  - Montmorilonite
  - Smectite
- Trioctahedral - Mg
  - Lizardite
  - Talc
  - Vermiculite
  - Biotite